MA798 Project: Randomized SVD for Basis Updating in POD-ROM

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Abstract

Reduced-order models (ROMs) are a common approach to perform complex analysis on high-fidelity models with prohibitive computational costs. Proper orthogonal decomposition-based ROMs (POD-ROMs), formulate a ROM by projection of high-fidelity model equations onto a reduced set of solutions called POD modes. Computation of POD modes is typically done by singular value decomposition (SVD), however, in many applications, such as large data dimension or data-streaming, randomized SVD and single view SVD can improve computation of POD modes. In this work, we assess the effect of randomized and single view SVD on accuracy of a POD-ROM applied to a liquid film model. We implement a greedy sampling algorithm to study if POD-ROM errors from alternative SVD algorithms propagate in POD-ROM optimization, leading to diverging models. We observe that the single view algorithm can have larger errors in POD reconstruction, but that the resulting POD-ROMs and greedy sampling optimizations are nearly identical.

1 Introduction

Proper orthogonal decomposition (POD) is a standard data reduction method which, forms a set of solutions to a dynamical system, termed snapshots, computes a set of orthogonal solutions, termed modes, spanning the same solution space. These orthogonal solutions are ordered according to the amount of variation in the original snapshots they quantify, allowing for easy data reduction by retaining only the most influential modes. For data from many dynamical systems, this can allow approximation of thousands of samples from tens of POD modes. An additional benefit of POD is that many reduced-order models (ROMs) can be formulated by assuming separation of variables, using the POD modes from the full-order model (FOM) as basis for spatial variation, and then solving for the temporal coefficients of the POD modes with a ROM.

Computation of POD modes is typically done via the singular value decomposition [3]. Consider a snapshot matrix $\mathbf{A} \in \mathbb{R}^{m\times n}$, where m is the number of discretization points and n is the number of snapshots with singular value decomposition $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$. Then the POD modes are the columns of $\mathbf{U}$ and the singular values are the amount of variation in $\mathbf{A}$ quantified by each mode. The reduced basis can be formed by truncating the modes to the the first p that together quantify a set proportion of information in the snapshot matrix, commonly 90%, 95%, or 99%. The SVD is an effective way to form and truncate a POD, however, the memory and computational costs of the SVD can make it intractable in some cases. Data from FOM simulations can very often be hundreds of gigabytes or more in size, particularly in 3D fluid flows. This large data size is complicated further when using multiple FOM simulations to quantify variations across a parameter space [4]. Additionally, computing the singular value decomposition for snapshot matrices at this scale can be exceedingly computationally expensive, especially since the vast majority of the SVD decomposition is removed in truncation [8]. Finally, in many applications, it can be advantageous to update an existing POD with new FOM data, particularly for greedy algorithms which use a POD or POD-ROM to identify optimal new samples [2]. Randomized SVD has been used to alleviate the computational and memory costs of the singular value decomposition, but still requires computation for the entire snapshot matrix and must be re-done to include any updated FOM data [5, 8, 7].

The single view randomized SVD algorithm has the potential to solve these issues by allowing updating of snapshot matrix approximation as new snapshot are added. It uses randomization to only compute the SVD on a reduced approximation, saving computation time. Iterative updating of the reduced approximation allows computing of the SVD for snapshots matrices too large to load on the given system. Finally, if new snapshots are added, randomized approximations of the row and column space only need to be recomputed for the new snapshots. In this work, we aim to quantify the accuracy and computational cost of a sequential learning POD-ROM using the single view

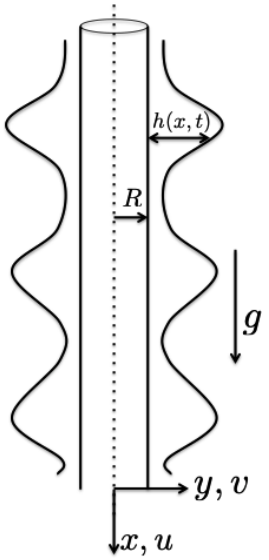


Figure 1: A figure from [6] depicting the thin film model. R is the radius and g represents gravity which works along the $x-$axis.

randomized SVD to form its POD basis. Error of the single view algorithm in computing the POD basis is already well described by expectation and large deviation bounds of error for the algorithm [9]. However, we seek to quantify how error in the POD may affect the accuracy of the resulting ROM. Additionally, we also seek to quantify to what extent error in the ROM will change FOM snapshot selection and accuracy for ROMs built using a greedy approach to iteratively update the POD basis.

2 Methodology

2.1 Liquid Film Model

To test the effectiveness of the single view randomized SVD for iterative basis formation, we will form POD-ROMs for a fluid flow problem. We consider the problem of a thin liquid film flowing down a vertical cylindrical fiber, shown in Figure 1. We assume radial symmetry, allowing this flow to be modeled by the one-dimensional PDE,

$$h_t + \frac{\partial}{\partial x}\left[h^3\left(\frac{1}{2} + h_x + h_{xxx}\right)\right] = 0, \tag{1}$$

where $h(x,t)$ is the film thickness [1]. We form the data matrix $\mathbf{A}$ by solving Equation (1) using a second-order finite difference solver with backwards Euler on the domain $(x,t) \in ([0,10] \times [0,3000])$ with $\Delta x = \dfrac{1}{160}$ and $\Delta t = 5$. We assume an initial condition of $h_0(x) = 0.355 + 0.01 \cdot \sin\left(\dfrac{\pi x}{10}\right)$ and periodic boundary conditions on h and its first three spatial derivatives.

2.2 POD-ROM Formulation

A common class of POD-ROMs are projection POD-ROMs. These models use solutions to a high-fidelity model, termed snapshots, and, assuming separation of variables, compute a reduced-set of orthogonal solutions in time and space, termed POD modes. The high-fidelity model equations can then be approximated with this decomposition and projected onto the span of the spatial modes to reduce all spatial differential operators to inner products between those operators and the spatial modes. This process reduces a PDE of space and time into an ODE for temporal coefficients which can be solved at a significantly reduced computational cost. Additionally, since the system is still

continuous in time and maintains the high-fidelity equations, the resulting POD-ROM can be solved at any new time and parameter value. However, we note that projection POD-ROMs are only accurate as long as the desired solution is within the span of the original spatial modes.

To compute the projection ROM for the PDE system,

$$h_t = -\frac{\partial}{\partial x} h^3(1/2 + h_x + h_{xxx}) = -\frac{3}{2}h^2 h_x - 3h^2 h_x^2 - 3h^2 h_x h_{xxx} - h^3 h_{xx} - h^3 h_{xxxx},$$

we first assume a POD decomposition for $h(t,x) \approx \sum_{i=1}^{k} a_i(t)\phi_i(x)$. For brevity, we note the use of the inner product $(\cdot,\ \cdot) = (\cdot,\ \cdot)_{L^2(0,10)}$. We then aim to solve the weak formulation with respect to the span of the spatial basis $\{\phi_i\}_{i=1}^k$, given by,

$$\begin{aligned}
-(h_t,\ \phi_i) &= (\frac{3}{2}h^2 h_x + 3h^2 h_x^2 + 3h^2 h_x h_{xxx} + h^3 h_{xx} + h^3 h_{xxxx},\ \phi_i),\\
&= \frac{3}{2}(h^2 h_x,\ \phi_i) + 3(h^2 h_x^2,\ \phi_i) + 3(h^2 h_x h_{xxx},\ \phi_i) + (h^3 h_{xx},\ \phi_i) + (h^3 h_{xxxx},\ \phi_i),\\
-\dot{a}_j(\phi_j,\ \phi_i) &\approx \frac{3}{2}a_j a_l a_m(\phi_j\phi_l\phi_m',\ \phi_i) + 3a_j a_l a_m a_n(\phi_j\phi_l\phi_m'\phi_n',\ \phi_i) + 3a_j a_l a_m a_n(\phi_j\phi_l\phi_m'\phi_n''',\ \phi_i)\\
&\quad + a_j a_l a_m a_n(\phi_j\phi_l\phi_m\phi_n'',\ \phi_i) + a_j a_l a_m a_n(\phi_j\phi_l\phi_m\phi_n'''',\ \phi_i)\\
\dot{a}_i &= -a_j a_l a_m(\frac{3}{2}\boldsymbol{L}^{(1)}_{ijlm} + a_n(3\boldsymbol{L}^{(2)}_{ijlmn} + 3\boldsymbol{L}^{(3)}_{ijlmn} + \boldsymbol{L}^{(4)}_{ijlmn} + \boldsymbol{L}^{(5)}_{ijlmn}))
\end{aligned}$$

such that,

$$\begin{aligned}
\boldsymbol{L}^{(1)}_{ijlm} &= (\phi_j\phi_l\phi_m',\ \phi_i) = \int_0^{10} (\phi_i(x)\phi_j(x)\phi_l(x)\phi_m'(x))\,dx \ \approx \sum_{p=1}^{N} (\phi_i(x_p)\phi_j(x_p)\phi_l(x_p)\phi_m'(x_p))\\
\boldsymbol{L}^{(2)}_{ijlmn} &= (\phi_j\phi_l\phi_m'\phi_n',\ \phi_i) = \int_0^{10} (\phi_i(x)\phi_j(x)\phi_l(x)\phi_m'(x)\phi_n'(x))\,dx \ \approx \sum_{p=1}^{N} (\phi_i(x_p)\phi_j(x_p)\phi_l(x_p)\phi_m'(x_p)\phi_n'(x_p))\\
\boldsymbol{L}^{(3)}_{ijlmn} &= (\phi_j\phi_l\phi_m'\phi_n''',\ \phi_i) = \int_0^{10} (\phi_i(x)\phi_j(x)\phi_l(x)\phi_m'(x)\phi_n'''(x))\,dx \ \approx \sum_{p=1}^{N} (\phi_i(x_p)\phi_j(x_p)\phi_l(x_p)\phi_m'(x_p)\phi_n'''(x_p))\\
\boldsymbol{L}^{(4)}_{ijlmn} &= (\phi_j\phi_l\phi_m\phi_n'',\ \phi_i) = \int_0^{10} (\phi_i(x)\phi_j(x)\phi_l(x)\phi_m(x)\phi_n''(x))\,dx \ \approx \sum_{p=1}^{N} (\phi_i(x_p)\phi_j(x_p)\phi_l(x_p)\phi_m(x_p)\phi_n''(x_p))\\
\boldsymbol{L}^{(5)}_{ijlmn} &= (\phi_j\phi_l\phi_m\phi_n'''',\ \phi_i) = \int_0^{10} (\phi_i(x)\phi_j(x)\phi_l(x)\phi_m(x)\phi_n''''(x))\,dx \ \approx \sum_{p=1}^{N} \phi_i(x_p)\phi_j(x_p)\phi_l(x_p)\phi_m(x_p)\phi_n''''(x_p)
\end{aligned}$$

A critical step of formation of an accurate POD-ROM is the computation of the POD modes. This is typically done via the singular-value decomposition, shown in Algorithm 1. However, the memory and computation costs of this algorithm can make it infeasible for many systems. In some circumstances, such as unstructured 3D fluid flows, the size of the data may be so large that it cannot be loaded onto a system. Additionally, the POD-ROM may require updating of its POD modes as new high-fidelity data is computed, such as in greedy sampling discussed in Section 2.4. In all of these cases, computation of POD modes using the standard SVD algorithm can be inefficient or impossible, but the streaming capabilities of the single-view algorithm could solve these limitations.

2.3 POD Algorithms

To reduce computational costs, the randomized SVD can be used to approximate the spatial and temporal POD modes, as shown in Algorithm 2. The randomized POD works by finding a matrix $\mathbf{Q} \in \mathbb{R}^{n\times(k+p)}$ with orthonormal columns such that range($\mathbf{Q}$) $\approx$ range($\mathbf{A}$) and computing the POD of $\mathbf{AQQ}^{\mathrm{T}}$. The addition of power iterations amplifies the dominant singular values and diminishes the non-dominant ones. Typically $q = 2$ is enough. This

Algorithm 1 Proper Orthogonal Decomposition

Input: $\mathbf{A} \in \mathbb{R}^{m\times n}$, target rank $1 \leq k \leq \text{rank}(\mathbf{A})$.
Output: Matrices $\mathbf{S} \in \mathbb{R}^{m\times k}$ and $\mathbf{T} \in \mathbb{R}^{n\times k}$ where columns of $\mathbf{S}$ and $\mathbf{T}$ correspond to discretizations of the k dominant spatial and temporal modes of respectively.

1: Compute rank-k truncated SVD of $\mathbf{A}$, $\mathbf{A}_k = \mathbf{U}_k\mathbf{\Sigma}_k\mathbf{V}_k^{\mathrm{T}}$
2: Let $\mathbf{S} = \mathbf{U}_k$ and $\mathbf{T} = \mathbf{V}_k\mathbf{\Sigma}_k$.

Algorithm 2 Randomized POD with Subspace Iterations

Input: $\mathbf{A} \in \mathbb{R}^{m\times n}$, target rank $k > 0$ and oversampling parameter $p \geq 0$ with with $k + p < \text{rank}(\mathbf{A})$, and number of subspace iterations $q \geq 0$.
Output: Matrices $\mathbf{S} \in \mathbb{R}^{m\times k}$ and $\mathbf{T} \in \mathbb{R}^{n\times k}$ where columns of $\mathbf{S}$ and $\mathbf{T}$ correspond to discretizations of the k dominant spatial and temporal modes respectively.

1: Draw a standard Gaussian matrix $\mathbf{\Omega} \in \mathbb{R}^{m\times(k+p)}$
2: Compute the $n \times (k + p)$ matrix $\mathbf{Y}_0 = \mathbf{A}^{\mathrm{T}}\mathbf{\Omega}$
3: Compute thin QR $\mathbf{Y}_0 = \mathbf{Q}_0\mathbf{R}_0$, with $\mathbf{Q}_0 \in \mathbb{R}^{n\times(k+p)}, \mathbf{R}_0 \in \mathbb{R}^{(k+p)\times(k+p)}$
4: **for** $j = 1, 2, \ldots, q$ **do**
5: Form the $m \times (k + p)$ matrix $\tilde{\mathbf{Y}}_j = \mathbf{A}\mathbf{Q}_{j-1}$
6: Compute thin QR of $\tilde{\mathbf{Y}}_j = \tilde{\mathbf{Q}}_j\tilde{\mathbf{R}}_j$, with $\tilde{\mathbf{Q}}_j \in \mathbb{R}^{m\times(k+p)}, \tilde{\mathbf{R}}_j \in \mathbb{R}^{(k+p)\times(k+p)}$
7: Form the $n \times (k + p)$ matrix $\mathbf{Y}_j = \mathbf{A}^{\mathrm{T}}\tilde{\mathbf{Q}}_\mathbf{j}$
8: Compute thin QR of $\mathbf{Y}_j = \mathbf{Q}_j\mathbf{R}_j$, with $\mathbf{Q}_j \in \mathbb{R}^{n\times(k+p)}, \mathbf{R}_j \in \mathbb{R}^{(k+p)\times(k+p)}$
9: **end for**
10: Form the $m \times (k + p)$ matrix $\mathbf{B} = \mathbf{A}\mathbf{Q}_q$
11: Obtain $\mathbf{S}$ and $\hat{\mathbf{T}}$ by performing Algorithm 1 on $\mathbf{B}$ with target rank k
12: Form the $n \times k$ matrix $\mathbf{T} = \mathbf{Q}_q\hat{\mathbf{T}}$

algorithm can be done computationally efficiently and may be feasible if the matrix $\mathbf{A}$ is too large for standard POD computations.

The other randomized method we consider is the single view algorithm for randomized POD. This method is beneficial if the matrix $\mathbf{A}$ is too large to fit in memory or if the data is streamed and the entire matrix is not available at once. This algorithm considers a matrix $\mathbf{A} = \begin{bmatrix}\mathbf{A}_0 & \mathbf{A}_1 & \ldots & \mathbf{A}_N\end{bmatrix}$ where $\mathbf{A}_i \in \mathbb{R}^{m\times n_i}$ and we can only work with one given $\mathbf{A}_i$ at a time. We begin by applying Algorithm 3 to $\mathbf{A}_0$ to obtain an approximate POD of $\mathbf{A}_0$. We oversample to have temporal dimension $\ell_1 = 2k + 1$ and spatial dimension $\ell_2 = 2\ell_1 + 1$. This oversampling is necessary for stability purposes. After we have taken the randomized POD of $\mathbf{A}_0$, when we have access to $\mathbf{A}_1 \in \mathbb{R}^{m\times n_1}$, we update the POD approximation to $\mathbf{A}_0$ by applying Algorithm 4 using $\mathbf{A}_1$ to obtain an approximate POD of $\begin{bmatrix}\mathbf{A}_0 & \mathbf{A}_1\end{bmatrix}$. We iterate this process, calling Algorithm 4 every time we obtain a new $\mathbf{A}_i$ to obtain a approximate POD of $\begin{bmatrix}\mathbf{A}_0 & \mathbf{A}_1 & \ldots & \mathbf{A}_i\end{bmatrix}$ until we've read in the entire matrix $\mathbf{A}$. In the next section, we introduce a greedy sampling approach where $\mathbf{A}_{i+1}$ is not known until we have the (potentially randomized) POD of $\mathbf{A}_i$. The single view algorithm gives a natural way to update the POD basis under this greedy sampling approach.

2.4 POD Updates with Greedy Sampling

A particular application of interest where updates to a POD is required is greedy sampling. In many cases where a ROM is applied, the high-fidelity model may be available to generate additional data, but too expensive to use for complex analysis. In these cases, it can be beneficial to use the chosen ROM to identify new design conditions to solve the high-fidelity model at, so as to quantify the most amount of information with the fewest samples. A particular approach for this, termed greedy sampling, is computing data from the high-fidelity model where the reduced-order model has the highest error [2]. This usually consists of iteratively solving the optimization problem,

$$\xi_i = \arg\max_{\xi} \|f_{HF}(\xi) - f_{ROM}(\xi; \xi_0, ..., \xi_{i-1})\|, \tag{2}$$

Algorithm 3 Single View Algorithm for Randomized POD

Input: $\mathbf{A} \in \mathbb{R}^{m \times n_0}$, target rank $k < \text{rank}(\mathbf{A})$.
Output: Matrices $\mathbf{S} \in \mathbb{R}^{m \times k}$ and $\mathbf{T} \in \mathbb{R}^{n_0 \times k}$ where columns of $\mathbf{S}$ and $\mathbf{T}$ correspond to discretizations of the k dominant spatial and temporal modes respectively, matrices $\mathbf{\Psi} \in \mathbb{R}^{(4k+3)\times m}$, $\mathbf{Y}_0 \in \mathbb{R}^{m \times (2k+1)}$, and $\mathbf{W}_0 \in \mathbb{R}^{(4k+3)\times n_0}$ used in future updates.

1: Draw a standard Gaussian matrix $\mathbf{\Omega} \in \mathbb{R}^{n_0 \times (2k+1)}$
2: Draw a standard Gaussian matrix $\mathbf{\Psi} \in \mathbb{R}^{(4k+3)\times m}$
3: Multiply $\mathbf{Y}_0 = \mathbf{A\Omega}$, with $\mathbf{Y} \in \mathbb{R}^{m \times (2k+1)}$
4: Multiply $\mathbf{W}_0 = \mathbf{\Psi A}$, with $\mathbf{W} \in \mathbb{R}^{(4k+3)\times n_0}$
5: Compute thin QR of $\mathbf{Y_0} = \mathbf{QR}$, with $\mathbf{Q} \in \mathbb{R}^{m \times (2k+1)}, \mathbf{R} \in \mathbb{R}^{(2k+1)\times(2k+1)}$
6: Multiply $\mathbf{X} = \mathbf{\Psi Q}$, where $\mathbf{X} \in \mathbb{R}^{(4k+3)\times(2k+1)}$
7: Solve for the $\mathbf{B} \in \mathbb{R}^{(2k+1)\times n_0}$ that satisfies the least squares problem $\min_{\mathbf{B}} \|\mathbf{XB} - \mathbf{W}_0\|_2$
8: Obtain $\hat{\mathbf{S}}$ and $\mathbf{T}$ by performing Algorithm 1 on $\mathbf{B}$ with target rank k
9: Form the $m \times k$ matrix $\mathbf{S} = \mathbf{Q}\hat{\mathbf{S}}$

where $f_{ROM}(\xi; \xi_0, ..., \xi_{i-1})$ is the ROM formed from high-fidelity data at design conditions $\xi_0, ..., \xi_{i-1}$.

Robust error analysis of randomized and single view algorithms already exists. However, two open questions is how does the error on POD-ROMs computed using those algorithms vary and how do any error variations propagate inside a greedy sampling algorithm where additional ROM error could change the design condition selected. To investigate this, we will apply a greedy sampling approach to snapshot selection for POD-ROMs applied to the thin film model described in Section 2.1. We will begin with a set of two snapshots, form their corresponding POD-ROM, solve the ROM over the entire time domain, add then add to the snapshots the two snapshots not used for the POD basis where the ROM has the highest error. We will repeat this until we observe an approximate convergence in the POD-ROM error. We will compare the accuracy and snapshots selected using SVD, randomized SVD, and single view. In particular, we will observe whether single view has significantly increased errors in the POD-ROM compared to SVD or randomized SVD, and observe if these errors lead to different snapshot selection and error propagation. This approach is particularly helpful since it avoids many common but difficult aspects of greedy sampling, such as optimization over a continuous design condition and solving of a high-cost model, but still quantifies the effect of different algorithms for updating POD in a greedy POD-ROM formulation.

3 Results

3.1 POD Results

We begin by comparing the effectiveness of the three different POD algorithms. We consider a matrix $\mathbf{A} = \begin{bmatrix}\mathbf{A}_0 & \mathbf{A}_1 & \mathbf{A}_2\end{bmatrix}$ where $\mathbf{A} \in \mathbb{R}^{1601\times 120}$ and $\mathbf{A}_i \in \mathbb{R}^{1601\times 40}$ for $i = 0, 1, 2$. We apply the POD, randomized POD with subspace iterations, and single view algorithm with $k = 2, p = 5$, and $q = 1$ for three matrices, $\mathbf{A}_0$ (40 snapshots of $\mathbf{A}$), $\begin{bmatrix}\mathbf{A}_0 & \mathbf{A}_1\end{bmatrix}$ (80 snapshots of $\mathbf{A}$), and then all 120 snapshots of $\mathbf{A}$. We reapply the full POD and randomized POD each time, and for single view, we apply the single view for $\mathbf{A}_0$ and then update it with $\mathbf{A_1}$ and $\mathbf{A}_2$. After we compute the PODs, we reconstruct the data at the time $t = 80$ and compare the reconstructions in Figure 2.

Since the reconstruction of the single view seems to vary greatly when snapshots are added, we increase the amount of oversampling in the single view method from $\ell_1 = 2k + 1 = 5$ and $\ell_2 = 2\ell_1 + 1 = 11$ to $\ell_1 = 3k + 1 = 7$ and $\ell_2 = 3\ell_1 + 1 = 22$. Figure 3 shows how increasing the oversampling affects the single view reconstruction.

3.2 POD-ROM Test Results

To test the POD-ROM, we computed a POD decomposition of all snapshots in $t \in [100, 3000]$ with $k = 6$. We left out the initial snapshots due to a short-time transience in these steps which POD-ROMs approximated poorly. The resulting POD quantified 99.6% of the variation within in the snapshots. Figure 5 shows the error of the resulting POD-ROM integrating from $t = 100$ to $t = 3000$. We observe that the error observes a slightly periodic behavior. The initial ROM error is quite low, but sharply increases to well-above 10% error, before decreasing at $t \approx 2,250$

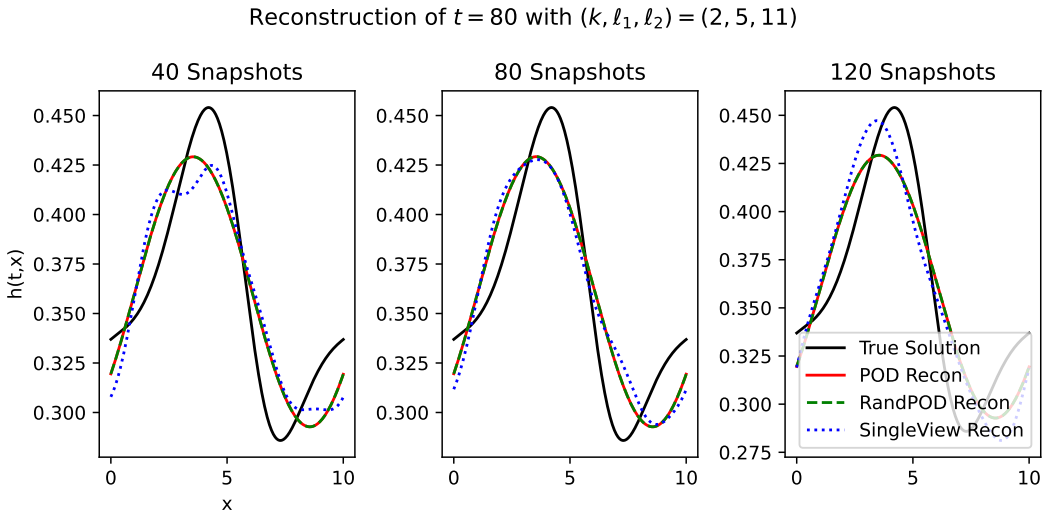


Figure 2: POD reconstructions of the first snapshot in a set with variable total snapshots and $(k, l_1, l_2) = (2, 5, 11)$. The reconstruction using Single View varies significantly as snapshots are added while POD and randomized POD stay consistent.

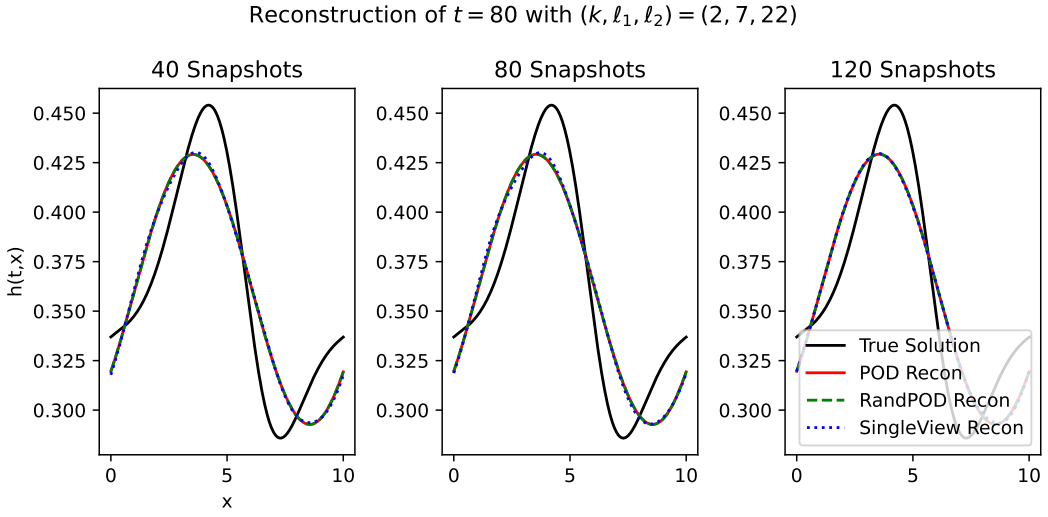


Figure 3: POD reconstructions of the first snapshot in a set with variable total snapshots and $(k, l_1, l_2) = (2, 7, 22)$. Increasing random sample dimension in Single View to $l_1 = 3 * k + 1$ and $l_2 = 3 * l_1 + 1$ gave equivalent results to POD and randomized POD.

Algorithm 4 Single View POD Update

Input: Matrix $\mathbf{A}_i \in \mathbb{R}^{m \times n_i}$, target rank $d < n$, matrices $\mathbf{\Psi}, \mathbf{Y}_{i-1}$, and $\mathbf{W}_{i-1}$ computed by Algorithm 3 or Algorithm 4

Output: Matrices $\mathbf{S} \in \mathbb{R}^{m \times k}$ and $\mathbf{T} \in \mathbb{R}^{\left(\sum_{j=1}^{i} n_j\right) \times k}$ where columns of $\mathbf{S}$ and $\mathbf{T}$ correspond to discretizations of the k dominant spatial and temporal modes of respectively, matrices $\mathbf{\Psi} \in \mathbb{R}^{(4k+3) \times m_0}$, $\mathbf{Y}_i \in \mathbb{R}^{m_0 \times (2k+1)}$, $\mathbf{W}_i \in \mathbb{R}^{(4k+3) \times \left(\sum_{j=1}^{i} n_j\right)}$ used for future updates.

1: Draw a standard Gaussian matrix $\mathbf{\Omega} \in \mathbb{R}^{n_i \times (2k+1)}$
2: Compute $\mathbf{Y}_i = \mathbf{Y}_{i-1} + \mathbf{A}_i \mathbf{\Omega}$, with $\mathbf{Y}_i \in \mathbb{R}^{m \times (2k+1)}$
3: Compute $\mathbf{W}_i = \begin{bmatrix} \mathbf{W}_{i-1} & \mathbf{\Psi} \mathbf{A}_i \end{bmatrix}$ with $\mathbf{W}_i \in \mathbb{R}^{(4k+3) \times \left(\sum_{j=1}^{i} n_j\right)}$
4: Compute thin QR of $\mathbf{Y}_i = \mathbf{QR}$, with $\mathbf{Q} \in \mathbb{R}^{m \times (2k+1)}, \mathbf{R} \in \mathbb{R}^{(2k+1) \times (2k+1)}$
5: Compute $\mathbf{X} = \mathbf{\Psi Q},$, where $\mathbf{X} \in \mathbb{R}^{(4k+3) \times (2k+1)}$
6: Solve for the $\mathbf{B} \in \mathbb{R}^{(2k+1) \times \left(\sum_{j=1}^{i} n_j\right)}$ that satisfies the least squares problem $\min_{\mathbf{B}} \|\mathbf{XB} - \mathbf{W}_0\|_2$
7: Obtain $\hat{\mathbf{S}}$ and $\mathbf{T}$ by performing Algorithm 1 on $\mathbf{B}$ with target rank k
8: Form the $m \times k$ matrix $\mathbf{S} = \mathbf{Q}\hat{\mathbf{S}}$

and almost reaching 1% at $t \approx 2750$, before increasing again. Though the relative error is quite high, periodicity in the true solution and the error indicates that high error may due the ROM solution being out of phase.

Figure 4 shows a comparison of the true temporal modes and the temporal modes computed by the POD-ROM for later time integrations. We computed the POD of all snapshots $t \in [100, 3000]$ with $k = 6$ and began solved for the modal coefficients with the POD-ROM starting at $t = 100$. We restrict the plot to $t \in [2100, 2700]$ to demonstrate that the POD-ROM accurately quantifies the periodic nature of the modal coefficients with the correct amplitudes, but at a slightly lower frequency. This causes the high error for many snapshots observed in Figure 5. We observe that the waves of the true and computed modal coefficients align at approximately $t = 2650$, consistent with the reduction in POD-ROM error observed at this time in Figure 5. We do observe a divergence between true and computed modal coefficients for Mode 1, but note that the relative magnitude of the divergence is extremely small, causing minimal error in the POD-ROM.

3.3 Greedy Optimization

The next step of our analysis was to test how error in ROM results may vary between different POD methods, and how those variations may propagate in a greedy sampling algorithm. To do this, we began by randomly sampling 2 snapshots from the snapshots in the time range $t \in [0, 3000]$. We then iteratively select new snapshots in batches of two for 15 iterations according to the greedy sampling approach we outlined in Section 2.4. We used $k = 4$ (or the number of snapshots when we used fewer than four) for all models. Figure 6 shows the maximum absolute relative error of all snapshots for each POD method and Greedy sampling iteration. We observe that the error is moderate for all cases, the POD-ROM produced using only two snapshots from all times achieved 23% error but adding 10 snapshots (iteration four) only reduced the error to 16% and further snapshots did not consistently lower the error, although slight increases and increases did occur. A difference between the POD algorithms only occurred after iteration 10, where single view had a slightly increased error compared to randomized and deterministic and selected different snapshots to add.

The greedy sampling results do indicate that single view can reliably be used to update a POD-ROM in greedy sampling applications. However, an important additional observation from these results is that models achieved almost the minimum error using only four snapshots, meaning that this dataset was actually a poor test of a greedy sampling approach since almost all variation exists in a small set of snapshots. This is likely due to the periodicity in solutions, causing many snapshots to be a near-exact copy of another. However, it is important to note that single-view began to be less accurate than randomized and deterministic as more snapshots were added, suggesting

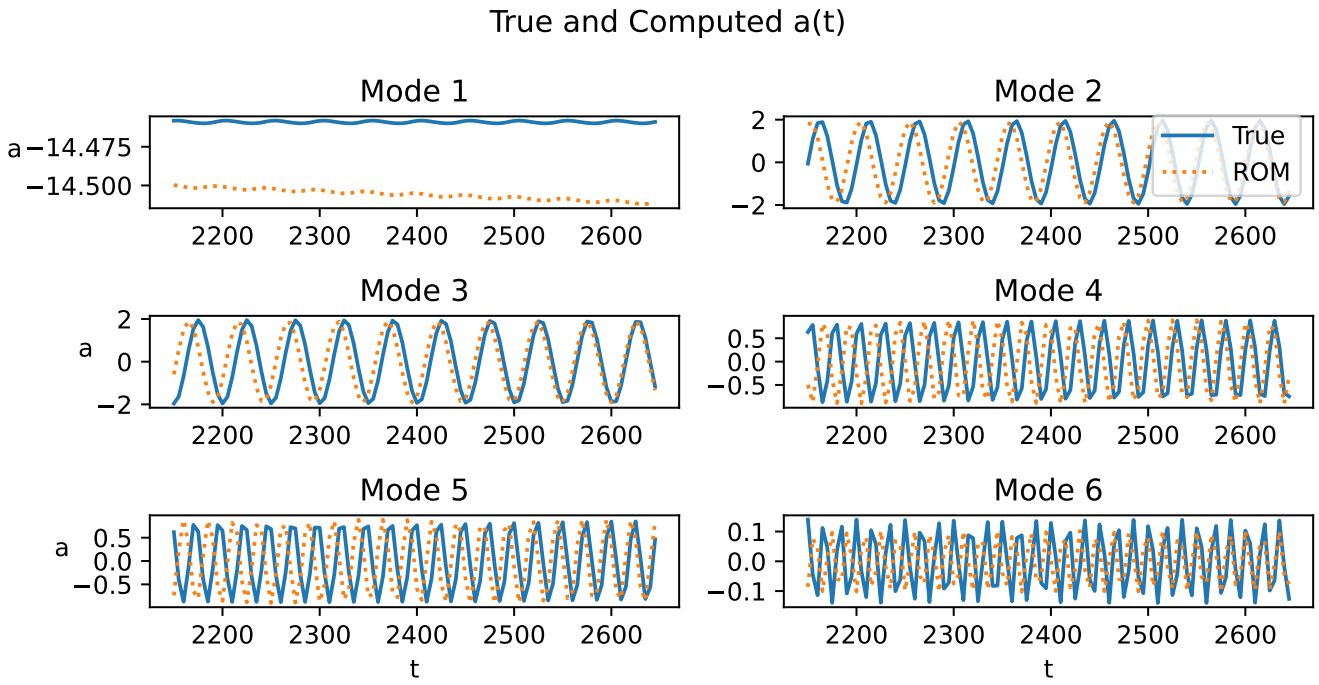


Figure 4: Comparison of true and ROM computed modal coefficients for $t \in [2100, 2700]$ for ROM integration starting at $t = 100$. The ROM accurately computes the magnitude of variations but at a slightly different frequency.

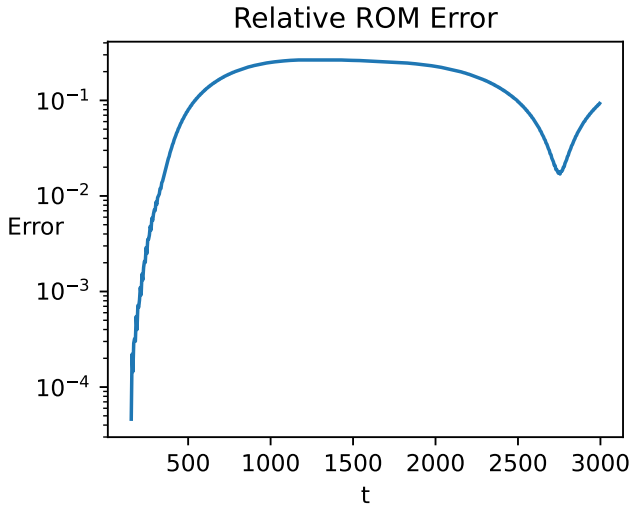


Figure 5: Relative error of ROM with $k = 4$ over entire temporal domain. The ROM quantifies early growth accurately but not the periodic phase, likely do to being out of phase.

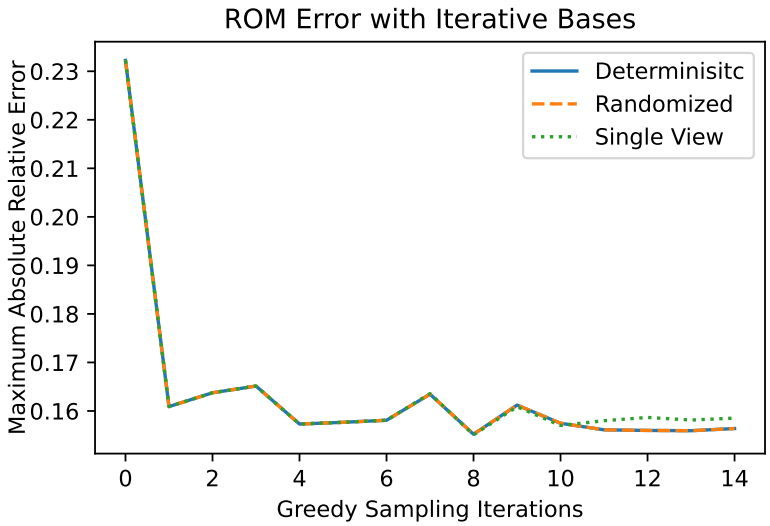


Figure 6: Greedy ROM generation results using $k = 4$, 3 initial snapshots, and batchsizes of 3. All POD approaches selected the same snapshots and achieved nearly the same error, despite significant data reduction in .

that single view could be significantly less accurate and lead to diverging solutions for more complex problems.

4 Conclusion

Proper orthogonal decomposition reduced order models (POD-ROMs) are a widely used tool for simplification of complex numerical models, particularly dynamical systems. POD-ROMs typically rely on the use of singular value decomposition (SVD) to compute the POD modes that are used in ROM development. However, SVD can be inefficient or intractable in cases with very large datasets or with iterative updating of data used to construct POD modes. To address these limitations, we compare SVD and randomized SVD to the single view algorithm which allows updating of the POD basis with new snapshots. The errors for each of these algorithms is well-defined by literature, so we instead investigate how those errors may propagate in a greedy sampling POD-ROM development in a liquid film test problem.

We found that at sufficient sampling for randomized SVD and single view, they were able to produce the same reconstruction as SVD, however, single view required an increase in the sampling dimension from the standard $l_1 = 2k + 1$ and $l_2 = 2l_1 + 1$ to $l_1 = 3k + 1$ and $l_2 = 3l_1 + 1$. The POD-ROMs made with a six mode decomposition, quantifying over 99% of data variation, were able to produce the periodicity and amplitude of modal coefficient variations, but slight differences in frequency led to persistent error for most time steps. Snapshot sampling via greedy sampling showed that all SVD methods identified the same snapshots and had the same error. After the POD basis was nearly optimal, single view had slight variation in snapshot selection and error, but not significantly.

These results demonstrated that the single view algorithm can be used as an alternative to SVD in POD-ROM development which saves computation time by not recomputing sketches for existing data. However, an important limitation of these results is that POD-ROMs using just four snapshots were able to accurately reconstruct the entire dataset, so effectively only one greedy sampling iteration was required. An important step for further analysis would be extension of this method to a test case with a greater degree of variation between snapshots so that further greedy sampling iterations may be required, allowing better confirmation that small differences between the SVD algorithms does not lead to diverging models.

References

[1] Biswal, S., Ji, H., Elamvazhuthi, K., and Bertozzi, A. L. Optimal boundary control of a model thin-film fiber coating model. *Physica D: Nonlinear Phenomena 457* (2024), 133942.

[2] Bui-Thanh, T., Willcox, K., and Ghattas, O. Parametric reduced-order models for probabilistic analysis of unsteady aerodynamic applications. *AIAA Journal 46*, 10 (2008), 2520–2529.

[3] Chatterjee, A. An introduction to the proper orthogonal decomposition. *Current Science 78*, 7 (2000), 808–817.

[4] Edwards, C., Lee, M. W., Ramezanian, D., and Smith, R. Quantifying emergent fluid dynamics using reynolds-interpolated fluid reduced-order models. In *AIAA SCITECH 2023 Forum* (2023).

[5] Halko, N., Martinsson, P.-G., and Tropp, J. A. Finding structure with randomness: Probabilistic algorithms for constructing approximate matrix decompositions. *SIAM review 53*, 2 (2011), 217–288.

[6] Ji, H., Falcon, C., Sadeghpour, A., Zeng, Z., Ju, Y. S., and Bertozzi, A. L. Dynamics of thin liquid films on vertical cylindrical fibres. *Journal of Fluid Mechanics 865* (2019), 303–327.

[7] Maulik, R., and Mengaldo, G. Pyparsvd: A streaming, distributed and randomized singular-value-decomposition library. In *2021 7th International Workshop on Data Analysis and Reduction for Big Scientific Data (DRBSD-7)* (2021), pp. 19–25.

[8] Ramezanian, D., Lee, M. W., and Moreno, I. B. Application of an affine nonlinear galerkin reduced-order model to compressible fluid flows. In *AIAA SCITECH 2023 Forum* (2023).

[9] Tropp, J. A., Yurtsever, A., Udell, M., and Cevher, V. Practical sketching algorithms for low-rank matrix approximation. *SIAM Journal on Matrix Analysis and Applications 38*, 4 (2017), 1454–1485.